

Milestone #: 3

Date: 2025.3.14

Group Number: 63

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By typing our names and student numbers in the above table, we certify that the work in the attached assignment was performed solely by those whose names and student IDs are included above. (In the case of Project Milestone 0, the main purpose of this page is for you to let us know your e-mail address, and then let us assign you to a TA for your project supervisor.)

In addition, we indicate that we are fully aware of the rules and consequences of plagiarism, as set forth by the Department of Computer Science and the University of British Columbia

Brief summary: Our project focuses on the database for management for rescued animals. It tracks animal details (species, rescue history), volunteer activities (schedules, specialized roles), adaptors, and medical care (vaccinations), while coordinating shelter resources (station capacity, donation funds).

Timeline and task breakdown/assignment:

1. Database Schema & SQL Script

- Finalize the ER diagram and relational schema. (Zeqian Feng, Zhitong Zou)

We need to check and modify the schema based on TA's feedback. This should be finished before the other steps.

- Implement a single SQL script based on the schema and put each entity with its SQL. (Yulin Yuan, Zhitong Zou)

This part is based on milestone 2, so this should be finished right after the feedback.

2. Query Implementation

- Implement the 10 required queries, ensuring:
 - Queries 1-6 are dynamic and accept user input.
 - Queries 7-10 are hardcoded. (Yulin Yuan, Zeqian Feng, Zhitong Zou)
- Queries to be implemented:
 - Insert (Yulin Yuan)
 - Update (Zhitong Zou)
 - Delete (Zeqian Feng)
 - Selection (Zeqian Feng)
 - Projection (Yulin Yuan)
 - Join (Yulin Yuan)
 - Aggregation with GROUP BY
 - Aggregation with HAVING
 - Nested Aggregation with GROUP BY
 - Division

We will finish this before 3.20.

3. GUI Development

- Implement a basic but functional GUI. (Yulin Yuan, Zhitong Zou)
- Ensure user-friendly navigation:
 - User should not enter raw SQL commands.
 - Use dropdowns, buttons, and forms for user input.
- Ensure the interface: (Zeqian Feng, Zhitong Zou)
 - Supports query execution.
 - Displays query results in a clean format.

We will finish this before 3.27.

4. Data Security & Error Handling

- Implement input sanitization to prevent SQL injection. (Yulin Yuan)
- Validate user inputs (Zhitong Zou)
- Provide error messages for:
 - Invalid input. (Zeqian Feng)
 - Foreign key constraint violations. (Zhitong Zou)
 - Duplicate entries. (Zhitong Zou)

We will finish this before 3.27.

5. Pre-Demo Preparation

- Ensure all files in the repository match the Milestone 4 submission (no post-deadline changes).
- Prepare a clean setup with: (Yulin Yuan, Zhitong Zou)
 - Database reset script.
 - Test data preloaded for queries.
- Plan the order of query demonstrations. (Zeqian Feng)
- Prepare examples for each query. (Yulin Yuan)
- Be ready to modify query inputs as requested by the TA.

We will finish this before 3.27.